

Md Tarif Chakder

Senior Android & Kotlin Multiplatform (KMP) Engineer

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📄 PROFESSIONAL SUMMARY

Senior Mobile Software Engineer with 7+ years of experience architecting, developing, and scaling mission-critical Android and cross-platform applications, with specialized domain expertise in **FinTech, Digital Wallets, Banking Integrations, and Android Smart POS Systems**. Deep technical mastery of modern Android development (**Kotlin, Jetpack Compose, Coroutines/Flow, Dagger-Hilt, Clean Architecture/MVVM**) and cross-platform delivery with **Kotlin Multiplatform (KMP)**. Proven track record implementing payment protocols (**ISO 8583, EMV**), terminal integrations (**Verifone, Ingenico**), biometric identity verification (**UAE PASS / Emirates ID**), and strict regulatory security standards (**PCI-DSS, DexGuard**). Active open-source library author and technical mentor.

🔗 CORE TECHNICAL COMPETENCIES

Languages & Core:	Kotlin, Java, C/C++ (NDK basics), SQL, Bash / Shell Scripting
Android & UI Engineering:	Jetpack Compose, Android SDK, Android Jetpack (Lifecycle, Room, Navigation, WorkManager), Material Design 3
Cross-Platform & Arch:	Kotlin Multiplatform (KMP), Compose Multiplatform (Android, iOS, Desktop JVM, Web), Clean Architecture, MVVM, MVI, Dagger-Hilt
FinTech, POS & Security:	Android Smart POS, ISO 8583 Messaging, EMV / NFC, Terminal SDKs (Verifone, Ingenico), UAE PASS / Emirates ID, PCI-DSS, DexGuard
Networking & DevOps:	gRPC, XMPP, SSL/TLS Sockets, Coroutines & Flow, Ktor Client, Retrofit, Git, CI/CD (GitHub Actions, Bitrise), Gradle Kotlin DSL

📁 PROFESSIONAL EXPERIENCE

Comera Pay Senior Software Engineer – Android	Abu Dhabi, UAE 2025 – Present
- Architect and develop flagship Android digital payment applications regulated by the Central Bank of the UAE (CBUAE), powering multi-currency VISA card issuance, digital wallets, and international remittances. - Implement highly secure end-to-end payment flows across Payment Gateways, Smart POS terminals, and dynamic QR-code payment rails, maintaining strict adherence to PCI-DSS compliance and cryptographic standards. - Integrate national digital identity verification via UAE PASS / Emirates ID, delivering streamlined biometric KYC verification, secure logins, and frictionless customer onboarding. - Spearhead the adoption of Kotlin Multiplatform (KMP) to share core business logic, cryptography, and network layers across platforms, accelerating feature delivery cycles by 35%. - Conduct comprehensive code reviews, enforce Clean Architecture and Jetpack Compose best practices, and mentor junior and mid-level engineers.	

- Spearheaded Android client architecture and feature development for **N-Genius**, an enterprise Smart POS payment solution handling in-person card transactions for merchants across the Middle East & Africa.
- Engineered low-latency, real-time device-to-server messaging protocols utilizing **gRPC** and **XMPP**, reducing transaction latency and ensuring zero packet loss during terminal handshakes.
- Integrated, tested, and certified payment kernels and SDKs across flagship hardware terminals (**Verifone** and **Ingenico**), supporting international schemes (Visa, Mastercard, Amex) and local regional cards.
- Led architectural refactoring of the core transaction flow into decoupled feature modules with MVVM, significantly improving testability and decreasing crash rates in live merchant deployments.
- Collaborated with hardware engineering, acquirer certification teams, and security auditors to achieve end-to-end payment gateway certification.

Geidea | Software Engineer – Android (POS & Payments)

Bangalore, India | 2022 – 2024

- Built next-generation Android POS financial applications for the Saudi Arabian (KSA) merchant acquiring ecosystem using **Jetpack Compose**, **Kotlin**, and **Dagger-Hilt**.
- Implemented full **ISO 8583 financial transaction packaging, parsing, and socket-level message transport**, ensuring rapid and resilient authorization with acquiring banks.
- Developed high-performance **POS communication libraries** using **SSL Sockets**, **Retrofit**, and **Kotlin Coroutines** for secure terminal-to-backend transaction handshakes.
- Configured and automated release pipelines integrating **DexGuard code obfuscation** and **binary tampering protection** within the CI/CD workflow to meet strict banking security mandates.

OMA Emirates Solution Gulf LLC | Software Engineer – Android

Dubai, UAE | 2022 – 2022

- Engineered customized Android Banking & Payment terminal applications for premier financial institutions including **National Bank of Oman (NBO)**.
- Authored reusable Android SDKs enabling reliable communication between POS hardware peripherals (contactless card readers, receipt printers) and banking client applications.
- Designed robust UI/UX transaction flows adhering to banking compliance and accessibility standards using Kotlin and MVVM.

🌐 OPEN-SOURCE INNOVATIONS & KMP LIBRARIES

MaterializeKMP **Kotlin Multiplatform** – Open-source library bringing Material Design 3 dynamic theming, styling, and essential UI components to Compose Multiplatform across Android, iOS, JVM Desktop, and Web.

Ktoast **Kotlin Multiplatform** – Unified, lightweight cross-platform Toast & notification framework built specifically for Kotlin Multiplatform projects to provide consistent alerts across platforms.

NetworkObserver **Kotlin Multiplatform** – Reactive Kotlin Multiplatform utility library for real-time network status monitoring and offline/online state management powered by Kotlin Coroutines & Flow.

PowerPrefs & ToastCompat **Android Utility** – Kotlin Android helper libraries delivering type-safe SharedPreferences wrappers and backwards-compatible UI toast mechanisms.

📖 PUBLISHED GOOGLE PLAY APPS

- **QR & Barcode Scanner:** CameraX & ML Kit based utility scanner.
- **Bengali Literary Series:** Cultural apps (*Nazrul*, *Tagore*, *Sarat Chandra*) featuring offline database caching.

🎓 EDUCATION

Maulana Abul Kalam Azad University of Technology (MAKAUT) 2015 – 2019
B.Tech in Computer Science & Engineering
(8.0 CGPA)